The Phenomenon: Why Occupancy-Grounded FB Fails on Manipulation, and How to Build an FB That Works

rescuing-fb — phenomenon note, 2026-07-20 (post T7/T8/T10 closure)

Factored-FB project

Status of the evidence. Every measure-side repair arm pre-registered in this program has now landed: T10-a (centered measure as auxiliary) 0.823 *— costs 8 points; T10-b2 (anchored measure-only)* 0.068/0.060/0.104 *— the historical ceiling; T7 (conservatism dose-response, collected 2026-07-20)* 0.240 *at exact-min and* 0.227 *beyond-min —* below *the plain-FB anchor* 0.42*; T8 (z-conditioned one-step FB, collected 2026-07-20)* 0.067 *— indistinguishable from its z-zeroed control* 0.058*. Per the pre-registered claim boundary (spec 04 §3), the strong negative claim is licensed. This note states the phenomenon that explains the entire record, then derives the constructive program from it.*

1 Setup and notation

MDP $(\mathcal{S}, \mathcal{A}, P, \gamma)$, $\gamma = 0.99$; offline dataset $\mathcal{D}$ of play episodes from a scripted pick-and-place policy μ (OGBench `cube-single-play`, 1M transitions); ρ the dataset state marginal. The successor measure of a policy π,

$$M^{\pi}(s, a, X) \;=\; \sum_{t \geq 0} \gamma^{t} \, \Pr\big[s_{t+1} \in X \mid s_0{=}s, a_0{=}a, \pi\big], \qquad X \subseteq \mathcal{S}, \tag{1}$$

is the object FB factorizes, $M^{\pi_z}(s, a, \mathrm{d}s') \approx F(s, a, z)^{\top} B(s') \rho(\mathrm{d}s')$, with the reward readout $Q_r(s, a) = F(s, a, z_r)^{\top} z_r$, $z_r = \mathbb{E}_{\rho}[r(s') B(s')]$. GC-IQL trains the same normalized bilinear head $Q(s, a, g) = \phi(s, a)^{\top} \psi(g) / \sqrt{d}$ by hindsight-relabelled sparse-reward expectile TD ($r \in \{-1, 0\}$, mask 0 at the goal). Both sides drive the identical FQL-style flow actor. Success numbers are OGBench `cube-single` 5-task means; anchors: bilinear GC-IQL **0.904**, plain FB **0.42** @1M, published GC-IQL 0.68.

2 The phenomenon, stated

Claim 1 (The phenomenon)**.** The successor-measure TD objective *is* scalar TD — performed simultaneously on the family of Dirac rewards — but with three estimator choices that, on play-distributed manipulation data, jointly destroy the action-ordering information the task requires: **(A)** it weights test tasks by state occupancy ρ rather than by the task distribution; **(B)** it bootstraps through a learned policy (non-contractive, and its only improvement mechanism) or, if made contractive (SARSA), loses improvement entirely; **(C)** it fits *revisit-counting occupancy* semantics rather than *first-passage* semantics, and under a re-grasping play policy the occupancy of

any test state is mixing-dominated: 82% of its γ-mass lies at horizons where futures are action-independent. Consequently the objective's sensitivity to the action taken now — the only thing the actor consumes — is orders of magnitude below its sensitivity to arm-configuration occupancy structure, its fixed point carries a near-vacuous advantage ordering, and *no conditioning, centering, scaling, conservatism, or bootstrap repair inside the occupancy corner can fix it*, because each such repair changes the optimization path toward the same fixed point. The bilinear structure, the z-space, and the moment inference API are all healthy; only the grounding is wrong. Replacing the grounding by task-weighted, in-sample, first-passage scalar TD — keeping every other FB component — restores control (0.825 zero-shot, inference gap 2.7 points).

The rest of this note makes each clause precise, checks it against all 12 constraint facts plus the newly landed T7/T8, and then derives the modifications that follow.

3 The bridge identity: measure TD is scalar TD on Dirac tasks

Proposition 1 (Exact rewriting of the FB loss)**.** *Let $m_{s^+}(s,a) := F(s,a,z)^\top B(s^+)$ denote the factorized measure density at test point s^+, and consider the implicit Dirac reward $r_{s^+}(s') = \delta_{s^+}(s')/\rho(s^+)$. Then the FB measure-TD loss with batch test points $s^+ \sim \rho$,*

$$\mathcal{L}_{\mathrm{FB}} = \mathbb{E}_{(s,a,s')\sim\mathcal{D},\, s^+\sim\rho}\Big[\big(m_{s^+}(s,a) - \gamma\,\mathrm{sg}\,\bar{m}_{s^+}(s',a')\big)^2\Big] - 2\,\mathbb{E}_{(s,a,s')}\big[m_{s'}(s,a)\big], \qquad a' \sim \pi_z(s'), \tag{2}$$

equals, up to an additive constant independent of the parameters, the ρ-weighted family of scalar TD losses

$$\mathbb{E}_{s^+\sim\rho}\,\mathbb{E}_{(s,a,s')}\Big[\big(m_{s^+}(s,a) - r_{s^+}(s') - \gamma\,\mathrm{sg}\,\bar{m}_{s^+}(s',a')\big)^2\Big]. \tag{3}$$

Proof. Expand the square in (3): the cross term $-2\,\mathbb{E}_{s^+\sim\rho}\mathbb{E}[(m_{s^+}(s,a) - \gamma\bar{m}_{s^+})\, r_{s^+}(s')]$ collapses under $\mathbb{E}_{s^+\sim\rho}[\, f(s^+)\delta_{s^+}(s')/\rho(s^+)\,] = f(s')$ to $-2\,\mathbb{E}[(m_{s'}(s,a) - \gamma\bar{m}_{s'})]$; the $\gamma\bar{m}$ part of it is a parameter-dependent term FB drops (target networks make it stop-gradient in practice), and $\mathbb{E}[r_{s^+}^2]$ is constant. What remains is exactly (2). □

So FB and GC-IQL train *the same bilinear object by the same kind of regression*, and differ only in four estimator coordinates:

Axis	FB (measure corner)	GC-IQL (working corner)	isolating evidence
A task weighting	test tasks $\sim \rho$ (occupancy)	hindsight mix (0.2/0.5/0.3)	§4; N1/N2 partial
B bootstrap	$a' \sim \pi_z$ (learned)	in-sample expectile	T8 + T7 (new, §5)
C value semantics	occupancy (counts revisits)	first-passage ($0/-1$ + mask)	§4; untested move
D conditioning of F	$\phi(s,a,z)$	$\phi(s,a)$, z at readout	fbz: -6 pts only

Corollary 1. *GC-IQL* is *FB after moving all four axes. "FB fails, GC-IQL works" is therefore not a comparison of two methods but the statement that some subset of $\{A,B,C\}$ is fatal (D is a 6-point detail). Every failed measure cell in the program — plain FB* 0.42*, measure-only* 0.08–0.34*, C4 rebuild* 0.165*, T10-b2* 0.06–0.10*, T7* 0.23–0.24*, T8* 0.067 *— shares* all three *of A, B, C at the FB corner. No cell has yet moved exactly one of A, B, C; §9 defines those cells.*

4 The quantitative core: temporal misalignment

The folk explanation — "play data is arm-dominated, so occupancies fit arm features" — is **false at the level of raw mass** and must not be printed: the cube is *held on 45.5% of dataset steps*, cube dimensions carry 26% of the observation's marginal variance, and per-step normalized motion share does not separate cube from antmaze in the needed direction (all measured model-free from the raw `npz`; misalignment note 2026-07-13). What is true is sharper and temporal.

Let $G_t \in \{\text{held}, \text{free}\}$ be the (observed, `obs[18]`) interaction gate and c_t the cube position. Measured on the raw dataset:

$$R_K := \frac{\mathbb{E}\big[\|c_{t+K} - c_t\| \;\big|\; G_t = \text{held}\big]}{\mathbb{E}\big[\|c_{t+K} - c_t\| \;\big|\; G_t = \text{free}\big]} = \begin{cases} 130\text{–}150\times & K \le 2 \\ 55\times & K = 5 \\ 7.8\times & K = 10 \\ 1.0\times & K \ge 50, \end{cases} \tag{4}$$

because the scripted play policy re-grasps: long-horizon occupancy *forgets the gate*. Against this, the discount kernel $\propto \gamma^t$ of (1) places 82% of its mass at $K > 20$.

Proposition 2 (Misalignment bound)**.** *Suppose the behavior-policy K-step kernels mix over actions:* $\sup_{a_1,a_2} \big\|P^\mu_K(\cdot \mid s, a_1) - P^\mu_K(\cdot \mid s, a_2)\big\|_{\mathrm{TV}} \le \varepsilon_K$ *with ε_K decaying as in (4). Then for any test function f with $\|f\|_\infty \le 1$ (any bounded reward or backward feature),*

$$\big|\, \langle M^\mu(s, a_1, \cdot) - M^\mu(s, a_2, \cdot), f \rangle \,\big| \;\le\; \sum_{t \le K_0} \gamma^t \, \varepsilon_t \;+\; \frac{\gamma^{K_0}}{1 - \gamma}\, \varepsilon_{K_0}, \tag{5}$$

i.e. the action-dependence of the occupancy object is confined to the pre-mixing window. With the measured profile ($K_0 \approx 20$–50), the second term — 82% of the total discounted mass — contributes essentially nothing: the occupancy functional of the data policy genuinely does not require the interaction variable at the horizons where its mass lives. *Control, by contrast, is decided at $K \le 5$, where the future is 55–150× mode-determined.*

Three corollaries, each independently verified:

1. *Fitting the measure well is insufficient.* The z-conditioned graft cut the measure's own TD residual 24× (6347→266) and moved success only to 0.255 (C7). The bound says why: the residual is dominated by the mixing-window mass, which is advantage-free; you can fit 99% of the objective while resolving none of the ordering.
2. *The domain flip.* On antmaze the short- and long-horizon conditional structure are the same smooth diffusion ($R_K \le 2$ at every K for every observable split tried) — there is no misalignment to pay, and measure-only scores 0.79 vs the 0.837 anchor there, against 0.10 vs 0.90 on cube (mdose).
3. *Warm starts cannot help.* The misalignment is a property of the *fixed point*, not the path: initialize at the converged 0.93 scalar-TD solution and fade $\mathcal{L}_{\mathrm{IQL}}$ out ⇒ success collapses to 0.05 in lockstep with the fade (iql-fade). Centering + scale anchors + split bootstrap (T10-b2) raised the early transient 3× (0.18–0.26 @50k) and then decayed to the same 0.06–0.10 ceiling: *path repaired, fixed point unchanged* — exactly what Claim 1 predicts for any within-corner repair.

Remark 1 (First-passage vs. occupancy, sharpened). Under first-passage semantics with absorption at the goal, the value is an affine function of $\mathbb{E}[\gamma^{T_{\text{hit}}}]$: dropping the cube delays T_{hit} by the re-acquisition time τ and multiplies the value by γ^{τ} — a persistent, non-washing penalty ($\gamma^{30} \approx 0.74$ per drop). Under occupancy semantics the same drop merely *shifts revisit mass later*; because the re-grasping data policy recovers the cube, the discounted deficit is second-order and the post-drop state is scored nearly as well as the held state. The 0/−1-with-mask convention GC-IQL trains *is* first-passage; this semantics was never transplanted onto the measure — it is the single cheapest untested axis move (§9).

Proposition 3 (Unsupported boundary)**.** *Of* 12,144 *grasp releases in the 1M-transition dataset,* 12,127 *are deliberate places and* 17 *are mid-air drops (*ρ*-mass* 1.7×10^{-5}*). For any objective that is an expectation under* ρ *(every objective in this program), relabelling those 17 transitions arbitrarily changes the loss by* $O(10^{-5}) \cdot L$*: the stable-vs-slipping value cliff — precisely the evaluation failure mode — is statistically invisible, and no bounded reweighting can amplify what is absent. Consequence: nothing may be asked to* rank grasp-stability counterfactuals*; the value's job is to order* in-support *grasp-maintaining actions, and the BC-anchored (flow / selection) extraction is load-bearing (bilinear+DDPG-BC* 0.51 < *bilinear+flow* 0.90*). Any proposal that loosens the BC anchor on cube is predicted to fail regardless of the critic.*

5 Axis B, resolved by T7 and T8 (new, 2026-07-20)

The two arms that landed today close axis B's two open corners.

T8: contraction without improvement is worthless. One-step FB (SARSA dataset-action bootstrap, a γ-contraction) with F kept z-conditioned scores 0.02/0.06/0.12 → **0.067**, statistically indistinguishable from its z-zeroed matched control (0.058, 10 seeds) — and from the BC-flow floor (≈ 0.06). This is the clean reading of the phenomenon: a contractive backup *estimates* M^{μ} faithfully, but M^{μ}'s advantage ordering is exactly the object Proposition 2 bounds — greedifying the behavior measure's readout adds *nothing* over BC on this data. One-step FB fails under both readings of the z-conditioning dispute; the ablation was worth running precisely because it kills both.

T7: the learned-policy bootstrap's optimism is not the bug — it is the only thing FB has. Exact min-aggregation (pessimism 0.5; verified ≡ min for twin heads) drops plain FB 0.42 → **0.240**; beyond-min (1.0) → 0.227. Conservatism *hurts*, dose-independently. Interpretation inside the phenomenon: plain FB's 0.42 > one-step's 0.067 because the $a' \sim \pi_z$ bootstrap is FB's sole improvement mechanism — it is what lifts FB off the BC floor at all — and pessimism shrinks the already-marginal advantage signal toward the ensemble's common (baseline) component. FB's pathology was never optimistic over-estimation of off-support actions (that is the DDPG+BC pathology of the *bilinear scalar* critic); it is an *uninformative ordering*, and you cannot fix a wrong ordering by shrinking it. This closes the VC-FB confound: the failure claim needs no conservative-FB restatement.

Corollary 2 (Axis B is a dilemma, not a knob)**.** *Within the measure corner, the bootstrap choice is: learned-*π *(improvement, no contraction, ceiling* 0.42*), SARSA (contraction, no improvement, floor* 0.067*), or the untested middle — an* in-sample expectile improvement on the measure *(§9, the exact transformation that took scalar GC-Bellman from unstable to* 0.904*). That cell is the only remaining way axis B could be exonerated or convicted.*

6 Scale identifiability (why the runaways were predictable)

The scalar objective's units are pinned by the reward: $|Q| \leq (1-\gamma)^{-1}$. The measure objective's units are density units $1/\rho$, and the factorization $F^\top B$ splits them non-identifiably ($F \to cF$, $B \to B/c$). Every observed runaway is this one defect: T10-b v1 (both legs scale-normalized $\Rightarrow$ objective scale-invariant $\Rightarrow$ diagonal attraction inflates q to 10^{13}), SG-ISFB's unbounded $\langle F, z\rangle$ drift, the $\psi(s,g)$ leg's norm runaway under scalar TD (T2: $\mathbb{E}\|\psi\|^2$ monotone in divergence). **Design rule (binding):** every future objective must answer "what pins the absolute scale?" before launch — computable rewards with units, a masked/bounded fixed point, or a hard norm anchor on exactly one leg.

7 The record against the phenomenon

Claim 1 is not one more hypothesis; it is the unique statement consistent with all of: C1–C12 (constraint set, Fable_Suggestions §0), the T10 closure (aux 0.823 — never helps, costs 8 points; anchored-only 0.06–0.10 — path not fixed point), T7 (0.240/0.227 — conservatism hurts), T8 (0.067 — contraction alone is the BC floor), the misalignment measurements (§4), the domain flip (antmaze 0.79), the in-distribution phase evidence (FB fails at C1 = 0.43 with grasp = 1.000 and $P(\text{success} \mid \text{secured}) = 0.437$ vs 0.982 — post-grasp ordering, not generalization), and the healthy-API results (E1a moment inference 0.955; RG-FB v0 0.825 ± 0.014 with a 2.7-point inference gap).

Claim 2 (Licensed negative claim — final wording)**.** On play-distributed manipulation data (OGBench cube-single), *no tested form of successor-measure TD grounding supports manipulation*: the native objective, its conditioned, normalized, centered, scale-anchored, conservative, one-step, auxiliary, curriculum, and warm-started variants all converge to a 0.06–0.42 band whose failures are post-grasp ordering failures, while the identical architecture, actor, data, and inference API under scalar first-passage TD reaches 0.90 (supervised) / 0.825 (zero-shot). The mechanism is the temporal occupancy–task misalignment of Proposition 2 (with the support boundary of Proposition 3); first-passage/hindsight-weighted measure objects (§9) are excluded from the claim's scope because they are no longer the successor measure.

8 Direction I — the constructive program (near-certain, run in this order)

The phenomenon dictates the repair: *keep FB-the-interface (bilinear factorization, z-space, moment inference, z-conditioned flow actor), replace FB-the-estimator with task-weighted in-sample first-passage scalar TD, and re-introduce FB's task-distribution training through the reward, not the measure.* Each rung below is decomposed into steps that are either theorems or already-measured numbers; the only genuinely new risk in each is named, with a guard.

8.1 GPI over goal candidates at inference (zero training; theorem-backed)

T6's sole zero-shot failure is composite rewards: two-goal max-mixtures score 0.260 while $\cos(z, \psi(g_k)) = 0.77$–$0.82$ — inference is fine; the actor has never seen an off-Dirac z. Classical GPI repairs this with no gradients. For $r = \max_k r_{g_k}$ with candidate set $Z = \{z_r, \psi(g_1), \ldots, \psi(g_K)\}$:

$$a_j = \pi(s, z_j) \;\; \forall j, \qquad \text{execute } a_{j^*}, \quad j^* = \arg\max_j Q(s, a_j, z_r). \tag{6}$$

Proposition 4 (GPI applies verbatim)**.** *$Q(\cdot,\cdot,z_r)$ is linear in z, so it is evaluated exactly where the critic was trained (a convex combination of Dirac readouts), and each candidate action is an on-support Dirac-policy action — the region where the critic's ordering is measured near-perfect (rerank* 0.91–0.96*). By the GPI theorem (Barreto et al.), the induced policy's value under r dominates every candidate's: $V_r^{\mathrm{GPI}}(s) \geq \max_j V_r^{\pi_{z_j}}(s)$ up to the critic's evaluation error on on-support actions. Since reaching* either *goal satisfies a max-mixture and per-goal success is* ≈ 0.9*, the prediction is* $0.260 \to \gtrsim 0.8$*.*

If GPI does *not* fix mixtures, the failure is not actor coverage — re-open the E1a ladder before touching training. Cost: $K{+}1$ critic passes per step, CPU, hours.

8.2 RG-FB at d=512 + selection extraction (both anchors already measured)

The v0 gap to the supervised anchor is training-side (0.825 vs 0.904; inference gap 2.7 pts). Two rungs with measured ceilings: (i) **d512 + z-actor**: the G4 anchor at d512 is 0.921 — v0 is d50; this is the known cheapest points. (ii) **IDQL-style selection at deployment**: $N{\approx}32$ BC-flow proposals, argmax frozen Q — the rerank instrument measured the bilinear critic's ordering worth 0.947–0.96 *with no actor training*, and selection is immune to both known extraction failure modes (off-support peak; anchor fights). Neither rung contains an untested mechanism.

8.3 RG-FB v1: task-prior scalar grounding (the headline combine)

FB's real asset was never the measure loss — it is that actor and critic are trained *across a distribution over tasks*. v0 trains both only at $z = \psi(g_{\text{hindsight}})$; T6 shows the cost (mixtures 0.260). v1 restores FB's task distribution through *computable rewards*, grounding every sampled task with the same expectile machinery: per batch, sample

$$\begin{aligned}
&\text{w.p. } p_1: \text{ Dirac goal (v0): } r(s') = \mathbf{1}[\|s'-g\| \leq \varepsilon], \quad z = \psi(g);\\
&\text{w.p. } p_2: \text{ } K\text{-goal mixture: } r(s') = \max_k \mathbf{1}[\|s'-g_k\| \leq \varepsilon], \quad z = \Sigma_k w_k \psi(g_k) \text{ (match-data renorm)};\\
&\text{w.p. } p_3: \text{ random linear reward: } r(s') = \mathrm{sg}[\psi(s')]^\top \tilde{z}/\sqrt{d}, \quad \tilde{z} \sim \text{sphere} \times \overline{\|\psi\|},
\end{aligned} \tag{7}$$

then apply GC-IQL's expectile value regression and scalar Bellman regression *at that* z, with the flow actor conditioned on $\mathrm{sg}[z]$. Nothing else changes.

Remark 2 (Why this is the principled combine). By Proposition 1, measure TD is scalar TD marginalized over Dirac tasks at ρ-weighting with a learned-π bootstrap. v1 is scalar TD marginalized over a *chosen* task prior with the in-sample bootstrap: it keeps the actual source of zero-shotness (training across tasks) and replaces every diagnosed pathology — axis A becomes task-shaped, axis B contractive-with-improvement, the scale is pinned because every sampled reward has units (§6), and the linear-in-z critic structure is *supervised* at off-Dirac z instead of assumed by extrapolation. The one new risk is multi-task interference on the Dirac anchor; guard with the pre-registered non-regression band (Dirac ≥ 0.80; two-goal max-mixture ≥ 0.60 success / ≤ 0.30 dead; regions ≥ 0.672).

8.4 The measure, kept — as a stop-gradient readout head

Do *not* re-add the measure loss at any λ on shared parameters (T10-a: -8 points; E2: at best free; the metamotivo aux-Q summed with the measure: 0.468, a null at 10 seeds). But two things about the measure survive intact and should be said and used:

Remark 3 (RG-FB does not discard the successor object — it swaps it). Under the 0/−1-with-mask convention, the trained value is an affine transform of the *first-passage kernel*:

$$\mathbb{E}\big[\gamma^{T_{\text{hit}}(g)}\big] \;=\; 1 + (1-\gamma)\, Q(s,a,g), \tag{8}$$

so $\phi^\top\psi/\sqrt{d}$ is a rank-d factorization of a bona fide successor-type object with first-passage (control-matched) semantics — Remark 1 at the level of the trained head. The SF-style extractables carry over: temporal distances $\hat{T}(s,g) = \log_\gamma\big(1+(1-\gamma)Q\big)$, reachability sets, linear goal-set composition + GPI (§8.1), and moment task inference. What is genuinely occupancy-specific — density queries, γ-model rollouts, OPE of dense rewards — is recovered below.

The sg readout head. Fit a full measure factorization *on top of* the frozen control towers,

$$\widehat{M}(s,a,s^+) \;=\; \phi_M\big(\text{sg}[\phi(s,a)]\big)^\top \psi_M\big(\text{sg}[\psi(s^+)]\big)/\sqrt{d_M}, \tag{9}$$

trained by the measure objective (or by direct Monte-Carlo occupancy regression) with *no gradient path into* ϕ, ψ. Every measured cost of "measure + scalar" arrived through shared parameters (E2 stop-grad localization; T10-a); cutting the sharing removes the conflict *by construction*, at zero control risk. This yields the honest measure fidelity number for the paper — evaluated against empirical discounted occupancy on held-out trajectories, broken down by horizon bucket ($K{\le}5$ / 5–20 / >20, so the misalignment bound of Prop. 2 is visible in the calibration) — plus working density/OPE extractables. The pre-registered comparison: if the head on scalar-trained features is at least as occupancy-faithful as plain FB's native $F^\top B$, the sentence is "scalar grounding is measure-*free* but not measure-*blind*"; if it is worse, print the trade — either way the control result is untouched. Implementation: spec 06, item 8.

9 Direction II — single-axis measure cells (mechanism work, optional for the claim)

With T7/T8/T10 dead, these cells no longer gate the negative claim (Claim 2 survives them semantically: a first-passage or hindsight-weighted object is not *the* successor measure). Their value is converting "all tested forms fail" into "axis X is the toxic one." Each is a small delta to the existing loss; prior on any single axis rescuing FB is low; run cheapest-diagnosis first.

9.1 Axis C: first-passage FB (one masking line)

Replace occupancy semantics with first-occupancy semantics ($F(s,a,s^+) = \mathbb{E}[\gamma^{T_{\text{hit}}(s^+)}]$): with batch test points, "the test point is hit" is a tolerance ball around the diagonal, $h_{ij} = \mathbf{1}[\|s'_i - s^+_j\| \le \varepsilon]$ (ε = the 4cm success radius on cube-xyz coordinates):

$$\delta_{ij} = M_{ij} - h_{ij} - \gamma\,(1-h_{ij})\,\text{sg}[\bar{M}_{ij}]. \tag{10}$$

Same fixed-point semantics as GC-IQL's masked sparse-reward value, per column; bilinear factorization, z-space, inference untouched; also removes the worst non-contraction amplifier (unbounded revisit mass). *Read:* $\gg 0.10 \Rightarrow$ semantics was a/the toxic axis (Remark 1 validated at the loss level); $\approx 0.10 \Rightarrow$ reframe the 82% number as correlate, not cause.

9.2 Axis B: expectile measure backup (the untested middle of Corollary 2)

Each test-point column of M is a scalar value function, so IQL applies pointwise: introduce $V_M(s, s^+) = \phi_V(s)^\top \tilde{\psi}(s^+)/\sqrt{d}$, train by expectile regression onto $\mathrm{sg}[M(s, a_{\text{data}}, s^+)]$, and back up

$$M(s, a, s^+) \leftarrow r_\delta + \gamma\, \mathrm{sg}[V_M(s', s^+)] \qquad (\text{no } a' \text{ of any kind}). \tag{11}$$

Contraction restored *and* improvement retained — the exact transformation that fixed the scalar side. Composes with (10); if singles are both partial, run C+B jointly as the "GC-IQL-semantics measure." *Read:* T8 ≈ 0.058 but this lifts $\Rightarrow$ "one-step evaluation is not enough; in-sample improvement is the missing bootstrap." Floors at $\sim 0.1 \Rightarrow$ axis B fully exonerated; A/C carry the blame.

9.3 Axis A: hindsight-weighted test points

Replace $s^+ \sim \rho$ with the scalar side's goal mix (50% geometric-future same-trajectory / 30% random / 20% current), moving the diagonal positive term to "a future state of this trajectory." This is the loss-level test of the weighting half of Proposition 2. (Caveat to print: at this corner the object is no longer the ρ-referenced measure without an importance correction $\rho_{\text{future}}/\rho$ — accept and say so.)

9.4 Do not resurrect (all refuted)

Centering/dueling splits of M (T10-b/b2); scale-normalizing both legs (runaway); measure as auxiliary at any λ *on shared parameters* (T10-a, E2; the sg head of §8.4 is the permitted form); the metamotivo aux-Q (`q_loss_coef`) summed with the measure (0.468 @1M, 10 seeds $\approx$ plain FB — it moves no estimator axis: self-referential implicit reward, learned-π bootstrap, no mask); curricula/annealing into a measure endpoint, *from any initialization including RG-FB* (iql-fade 5.3; shared-legs ramp 0.9; sg-ϕ ramp 84.9 = neutralized-not-helpful: the fixed point owns the endpoint, warm starts only change the path); z-zeroed one-step (0.058); **conservatism/min-aggregation at any dose (T7, new)**; eval-time goal smoothing (T1); unbounded $\psi(s, g)$ heads (T2); task-norm reweighting (N1/N2); zHER / decouple-actor-z; whitened inference ($\kappa \approx 10^{14}$).

10 Certainty budget (what "almost 100%" means here)

Proposal	theorem part	measured part	new risk (guard)
GPI (§8.1)	GPI dominance; Q linear in z	ordering 0.91–0.96; per-goal 0.9	eval-error off-Dirac (report per-pair)
d512 rerun (§8.2)	—	G4 anchor 0.921	seed noise (3 seeds)
selection (§8.2)	one-step/IDQL optimality	rerank 0.947–0.96	N-sweep peak drift (cap N=32)
v1 task prior (§8.3)	Prop. 1; scale pinned	v0 0.825; E1a 0.955; T6 regions 0.672	interference (non-regression band)
first-passage (§9)	Remark 1	—	low prior; diagnostic value only

The Direction-I stack has no untested mechanism anywhere in its chain; its composite risks are interference effects, each with a pre-registered guard band. That is the strongest form of "works with near-certainty" an empirical program can honestly claim — every link is either a theorem, an already-measured number, or a guarded regression test.

Implementations for every item in §8–9 are specified knob-by-knob in `specs/06-phenomenon-implementations.md`. *Companion document for research question 1 (why the bilinear form works):* `PAPER/ogbench-sota/phenomenon.tex`.